DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS

COURSE CODE	**:**	S-ITNT416LA
COURSE TITLE	**:**	Big Data And Analytics
COURSE TYPE	**:**	Laboratory
COURSE CREDIT	**:**	1 Unit
LEARNING MODALITY	**:**	Onsite
PRE-REQUISITES	**:**	S-ITNT344LC
CO-REQUISITES	**:**	S-ITNT416LC
PROFESSOR	**:**	Paulino H. Gatpandan, DIT phgatpandan@dlsud.edu.ph
CONSULTATION HOURS	**:**	1000 – 1200 Tuesday (COS100C) 1000 – 1200 Wednesday (COS100C) 0900 – 1200 & 1300 – 1430 Friday (Online)

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

Provides students with an overview of the current trends in information technology that drives today's business. The course provides understanding on data management techniques that can help an organization achieve its business goals and address operational challenges. This also introduces different tools and methods used in business analytics to provide the students with opportunities to apply these techniques in simulations in a computer laboratory.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO1. Apply data transformation techniques including standardization and encoding of categorical data using Pandas and NumPy.

CLO2. Create data visualizations and dashboards to communicate analytical results

effectively.

CLO3. Analyze datasets using machine learning algorithms and tools to extract meaningful insights.

CLO4. Explain the purpose and activities involved in each stage of the life cycle.

CLO5. Demonstrate 21st century skills in all learning activities.

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO 1. Apply data transformation techniques including standardization and encoding of categorical data using Pandas and NumPy.	TLO1. Differentiate between various data transformation techniques such as standardization, normalization, and encoding. TLO2. Explain the need for encoding categorical data and the different types of encoding techniques.
CLO 2. Create data visualizations and dashboards to communicate analytical results effectively.	TLO3. Demonstrate proficiency in using data visualization tools. TLO4. Design clear and informative visualizations that accurately represent data insights.
CLO 3. Analyze datasets using machine learning algorithms and tools to extract meaningful insights.	TLO5. Preprocess datasets by cleaning, normalizing, and transforming data to be suitable for machine learning models. TLO6. Demonstrate the practical application of machine learning models to derive actionable insights from various datasets.
CLO 5. Demonstrate 21st century skills in all learning activities.	TLO8. Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)

Module Number and Learning Outcome/s	Date	Teaching-Learning Lessons and Activities / Assessment Strategies	Technology Enabler	Onsite	Asynchronous	Allotted Hours
Introduction Module **Module 1** CLO1 CLO5 TLO1 TLO2 TLO9	Week 1-3 Aug 19 - Sep 7	**Course Orientation** • Gospel Reading, Reflection and Prayer • Course Overview • Course Polices and Requirements • Professor Profile • Grading System **Data Generation and Acquisition** • Data Sources and Types • Data Acquisition and Methods • Data Quality and Preprocessing • **Enabling Assessment:** • Identify and analyze data sources for potential insights • Laboratory activity on data preparation: cleansing and transformation	Schoolbook Power Point	/		9.0
Module 2 CLO2 CLO5 TLO3 TLO4 TLO9	Week 4-5 Sep 9-21	**Data Analysis and Visualization** Gospel Reading, Reflection and Prayer • Exploratory Data Analysis • Data Visualization • Tools for Data Visualization • **Enabling Assessment:** • Laboratory activity on data analysis • Laboratory activity on data visualization	Schoolbook Power Point	/		6.0

Module 3 CLO3 CLO5 TLO5 TLO6 TLO9	Week 6-7 Sep 23 - Oct 5	**Machine Learning with Big Data** • Gospel Reading, Reflection and Prayer • Introduction to Machine Learning • Machine Learning with Big Data • **Enabling Assessment:** • Laboratory activity on Correlation Analysis • Laboratory activity on Regression	Schoolbook Power Point	/		6.0
	Week 8 Oct 7-12	**Summative Assessment** Midterm Examination		/		3.0
			Total	**24.0**		**24.0**

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO 3. Explain the purpose and activities involved in each stage of the life cycle.	TLO5. Preprocess datasets by cleaning, normalizing, and transforming data to be suitable for machine learning models. TLO6. Demonstrate the practical application of machine learning models to derive actionable insights from various datasets.
CLO 4. Demonstrate how big data analytics drives decision-making processes in different industry contexts.	TLO7. Identify and describe major applications of big data across various industries such as healthcare, finance, retail, manufacturing, and telecommunications. TLO8. Explain how big data solutions address specific industry challenges and opportunities.
CLO 5. Demonstrate 21st century skills in all learning activities.	TLO9. Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)

Module Number and Learning Outcome/s	Date	Teaching-Learning Lessons and Activities / Assessment Strategies	Technology Enabler	Onsite	Asynchronous	Allotted Hours
Module 3 CLO3 CLO5 TLO5 TLO6 TLO9	Week 10 Oct 21-26	**Machine Learning with Big Data** (continuation) • Gospel Reading, Reflection and Prayer • **Enabling Assessment:** • Laboratory activity on Classification	Schoolbook Power Point	/		3.0
	Week 11 Oct 28 – Nov 2	Independent Learning Week	Schoolbook		/	3.0
Module 3 CLO3 CLO5 TLO5 TLO6 TLO9	Week 12-13 Nov 4-16	**Machine Learning with Big Data** (continuation) • Gospel Reading, Reflection and Prayer • **Enabling Assessment:** • Laboratory activity on Clustering	Schoolbook Power Point	/		6.0
Module 4 CLO4 CLO5 TLO7 TLO8 TLO9	Week 14-16 Nov 18 - Dec 7	**Industry Applications of Big Data** • Gospel Reading, Reflection and Prayer • Big Data in Healthcare • Big Data in Finance • Big Data in Retail • Enabling Assessment - Case Studies	Power Point	/		9.0
	Week 17 Dec 9-11	**Summative Assessment** Final Examination		/		3.0
			Total	**21.0**	**3.0**	**24.0**

GRADING SYSTEM:

Midterm		**Finals**	
Enabling Assessments	– 40%	Enabling Assessments	– 40%
Summative Assessments	– 60%	Summative Assessments	– 60%
	100%		100%

Overall Final Grade = (Midterm + Finals) / 2

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Onsite Learning Mode. All activities will be done in the laboratory classroom and field.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar.

3. Laboratory Policies
   a. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
   b. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless special permission is sought. Cellular phones also cannot be used as calculator during examination.
   c. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

4. Communication and Feedback
   a. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
      i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
      ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
      iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
   b. No use of foul, curse, and other bad words, whether written or verbal. No shouting. No flooding of chat and messages. No selling.

5. Attendance Policy
    a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for student's grade.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

6. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

7. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.

REFERENCES:

Call number or e-provider	References
HD 69.B87.M184 2023	Hemachandran K, et al. 2023. Machine learning for business analytics: real-time data analysis for decision-making. New York: Routledge
QA 76.9.B45 .C378 2021	Castellani B, Rajaram R. 2021. Big data mining and complexity. London: SAGE Publications
HD 30.27 .M299 2019	Malviya A, Malmgren M. 2019. Big data for managers: creating value. London; New York: Routledge, Taylor and Francis group
QA 76.9.D26 .C816 2019	Coronel C, Morris S. 2019. Database systems: design, implementation and management. 13th edition. (Australia, United States): Cengage Learning

Prepared: **Information Technology Department**
1st Semester 2024-2025

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Chair, Information Technology Department

Endorsed: **(SGD) Ms. Marivic R. Mitschek**
Dean, College of Information and Computer Studies

cf: OVCAR, AEA, CICS Curriculum Development Committee, file